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Ecology is not yet ready for AI—and why that matters

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Camera traps, such as this one that captured an image of a leopard cat in a Thailand forest, exemplify how researchers use AI-enabled tools to monitor ecological systems. Image credit: Shutterstock/forest71.

Artificial intelligence (AI) is rapidly becoming integral to ecological research. Deep learning models can, for example, identify species from camera trap images, including large mammals such as deer, elephants, and carnivores, and often achieve accuracies exceeding 90% while dramatically reducing the time required for manual annotation. Neural networks can sift satellite imagery to detect forest cover change and classify vegetation types across large spatial scales, enabling biodiversity monitoring at resolutions and speeds that were previously unattainable (1).

In a world experiencing accelerating environmental change, ecologists understandably view AI as a necessary extension of analytical capacity rather than an optional tool. The scale, speed, and complexity of ecological data increasingly exceed what traditional methods can handle. In that sense, AI is not simply useful; it is becoming essential. But necessity does not imply readiness. Despite clear successes, the growing presence of AI in ecology masks a deeper problem: The field is not yet fully prepared to integrate these tools responsibly or interpret their outputs reliably across all contexts. We need to rethink our use of the technology and to demand a higher degree of caution, contextual understanding, and domain expertise.

What's Special About Ecology

While similar challenges exist in other scientific fields, they are often amplified in ecology. Unlike controlled experimental systems in molecular or cell biology, for example, ecology relies heavily on heterogeneous, observational data collected

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across variable spatial and temporal scales. AI can struggle in such settings, which makes it unsuitable for many core ecological applications, including species distribution modeling, biodiversity forecasting, and ecosystem prediction. In these domains, limitations in data quality, ecological complexity, and interpretability become far more pronounced.

Ecological data also rely heavily on field observations and reflect historical contingencies, uneven sampling, and methodological variation. Species occurrence records contain false absences, biodiversity databases are taxonomically biased, and long-term datasets are often fragmented by logistical and funding constraints.

When machine learning models are trained on such data, they can produce outputs that appear statistically robust, yet lack ecological plausibility. While such errors are not unique to AI and can arise in traditional modeling approaches as well, AI can amplify them by identifying statistical patterns without adequately accounting for underlying ecological constraints, such as dispersal limits, physical barriers, and species interactions. For example, distribution models have falsely predicted the presence of species in regions that are inaccessible due to dispersal barriers. They suggest, for example, that Alpine plants will somehow spread to distant mountain ranges or enable trees to cross the Amazon river. Others interpret absence records as evidence of ecological exclusion when they actually reflect sampling gaps.

The issue is not whether ecology should use AI, but how to approach it with greater transparency, critical evaluation, and ecological grounding.

These are not trivial errors. They illustrate a central tension: AI can detect patterns in ecological data without understanding the processes that generate them.

Ecosystems are also dynamic systems shaped by shifting baselines, time lags, ecological memory, and nonlinear feedback (2–4). Climate change, land-use shifts, and extreme events continuously alter ecological relationships. Many AI models, however, implicitly assume that relationships remain stable over time, that training data are representative of future conditions, and that observations are independent. These assumptions are routinely violated in ecological systems. As a result, models trained on historical data may fail under novel conditions, producing confident predictions that mislead and skew decision-making (5). Ecology has always relied on past observations to understand the future—but in rapidly changing systems, this reliance becomes a liability. The future increasingly falls outside the range of the past (1–6).

In this context, predictive accuracy alone is not sufficient. A model may perform well statistically while still failing to capture underlying ecological processes or produce meaningful insights for conservation and management.

Adapting to Technology

Ecology has faced similar moments before. The rise of neural networks in the late 20th century and the expansion of web-based ecological data in the early 2000s both promised analytical breakthroughs. In each case, enthusiasm initially outpaced the development of shared standards, validation frameworks, and interpretive norms.

Take the development of climate and ecosystem models. Advances in modeling capacity surged ahead of systematic approaches to uncertainty quantification and validation, leading to debates about reliability and interpretation (5). Over time, the field adapted—not by rejecting these tools, but by developing the conceptual and methodological frameworks needed to use them responsibly.

A similar dynamic is now unfolding with AI. Its adoption is accelerating faster than the development of shared standards for interpretation, validation, and governance. But the difference now is that AI is a more powerful, more opaque, and more widely accessible tool—in ecology and any number of other fields—which raises the stakes considerably.

There is an added risk that wide adoption of AI could reinforce existing inequities, particularly when data generated by global research networks and Indigenous communities are analyzed elsewhere without adequate governance, credit, or data sovereignty. Effective AI applications depend on interoperable datasets, sustained data curation, and access to computational resources. But these remain unevenly distributed. Ecological data are often siloed across institutions and nations, while high-performance computing is concentrated in a relatively small number of well-resourced laboratories. This pattern predates AI, but AI may intensify the trend, thereby increasing the value of centralized data processing and further concentrating analytical power.

The most immediate constraint is education. Most ecologists lack formal training in machine learning, algorithmic bias, and uncertainty interpretation, while many AI practitioners lack ecological intuition. Interpreting ecological data requires a deep understanding of system dynamics, historical context, and process-based reasoning. Without that grounding, models may be technically sophisticated, yet ecologically fragile—producing outputs that are difficult to interpret or easy to misinterpret. All of this means that AI integration risks remaining superficial: powerful tools applied without sufficient understanding of their limitations or ecological meaning.

Using AI Well

This is not an argument against AI in ecology. On the contrary, AI is urgently needed. It can enhance biodiversity monitoring, improve conservation planning, detect early warning signals of ecosystem collapse, and integrate complex datasets in ways that exceed human capacity.

The issue is not whether ecology should use AI, but how to approach it with greater transparency, critical evaluation, and ecological grounding. For example, when an ecologist uses camera trap data to train an AI model, failing to account for sampling bias or detection limitations can produce misleading estimates of species presence or abundance, underscoring the need for greater transparency, careful validation, and ecologically grounded interpretation. This will require coordinated action across the ecological community. Journals and reviewers should require transparency in model assumptions, validation procedures, and ecological interpretation—not just predictive performance.

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Funding agencies should support interdisciplinary training and require attention to ethical considerations, data governance, and reproducibility challenges that extend beyond ecology and are increasingly relevant across fields adopting AI, including areas such as genomics, climate science, and epidemiology. Universities should integrate AI literacy into ecological education at undergraduate and postgraduate levels. And researchers must engage critically with AI tools, grounding their use in ecological theory and domain knowledge.

The field now faces a choice. It can adopt these tools rapidly and risk producing results that are technically impressive, but ecologically misleading. Or it can invest in the slower, more difficult work of building the conceptual, infrastructural, and ethical foundations needed to use AI well.

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